

---

---

---

---

---



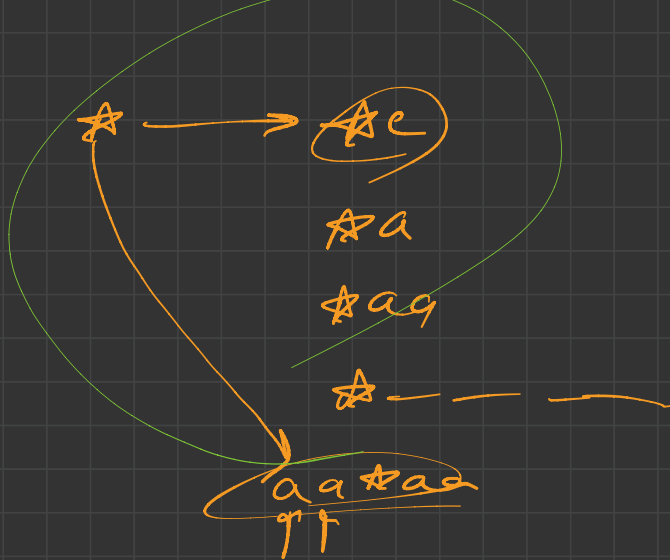
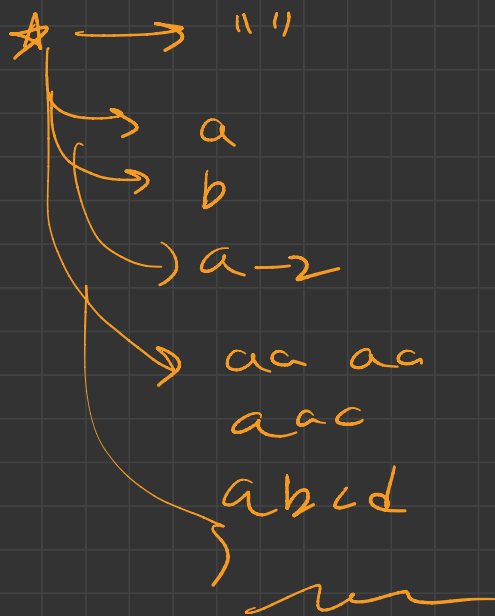
# Dynamic Programming

## → Wildcard Pattern Matching

✓ str = "a b c d e"  
✓ pattern = "a \* c ? e"  
↓  
b      d  
Valid pattern

? → matches with any single character

\* → matches any sequence of character



$s = \underline{\text{"a|a"}}$   
 $p = \underline{\text{"|a"}}$  → Invalid pattern

$s = \underline{\text{"abc"}}$   
 $p = \underline{\text{"a|b|c"}}$  → Valid pattern

$s = \underline{\text{"ab"}}$   
 $p = \underline{\text{"a|a|a"}}$  → Invalid pattern

str = " a b c d e "  
pattern = " a \* c P e "

Diagram showing indices  $i$  and  $j$  pointing to the characters 'e' in both strings.

match  
 $\rightarrow str[i] == pattern[j]$   
 $\rightarrow pattern[j] == '?'$

if (match,  
 $\rightarrow f(str, pattern, i-1, j-1);$

else if (pattern[j] == '\*')

{ return  $f(str, pattern, i, j-1) \rightarrow \text{Eis}$

or  $f(str, pattern, i-1, j) \rightarrow * \rightarrow *$

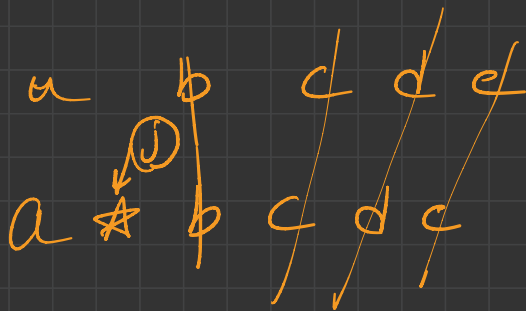
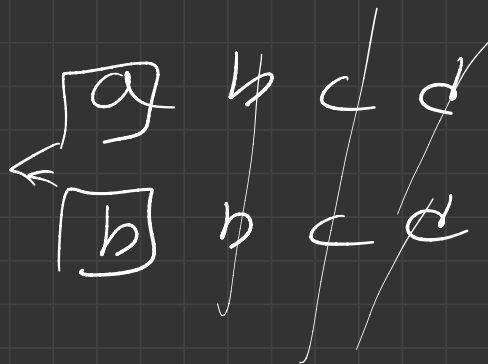
$f(str, pattern, i, j)$

$*$   $\rightarrow$  empty string

Koi bhi

else

return false;



$s \rightarrow \underline{a} \underline{b} c \mid d c$   
 $p \rightarrow \boxed{c ? c}$

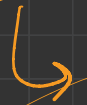
$s \rightarrow \underline{a} b c$   
 $p \rightarrow \boxed{b} a b c$   
 $\swarrow$   
 $\underline{T.P}$

$s \rightarrow \underline{a} b c$   
 $p \rightarrow \boxed{\star} a b c$   
 $\downarrow$   
 $E.S$   
 $\searrow$   
 $\bigcirc V.P$

B.V

Rec Mem  
B.V  
S.O

T.D

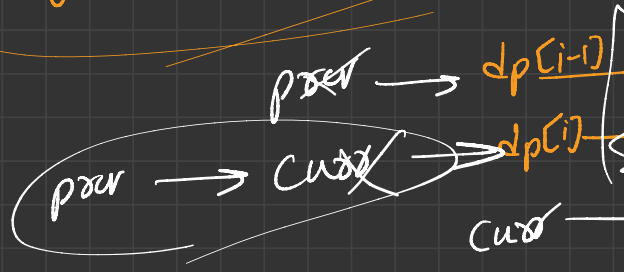


$i \rightarrow 0 \rightarrow s.len$

$0 \rightarrow 0 \rightarrow p.len$

$T.C \rightarrow O(n \times m)$   
 $i \rightarrow s.len \rightarrow 0$   
 $j \rightarrow p.len \rightarrow 0$

$S.C \rightarrow O(m)$



after each itr  
 $prev = curr$

